

TLH Sleeve Architecture

TLH SLEEVE — ARCHITECTURE & INFRASTRUCTURE DESIGN

SignalOS desk · 2026-08-26 · plan-of-record synthesis (runbook 2026-08-21 + amendments) with build inventory and open decisions

1. Executive summary

The tax-loss-harvesting sleeve exists to do one thing on one deadline: **realize enough capital losses by 2026-12-31 to offset the ~\$5.0M zero-basis gain landing in September** (~37.1% combined marginal — every harvested dollar saves ~\$0.37 against *this year’s* bill; anything harvested in 2027 is merely a carryforward). Everything else — account structure, basket mechanics, hedges, and this week’s order-integrity infrastructure — is in service of that constraint plus one boundary: **the harvest machine must never contaminate the judgment machine.**

Design in one paragraph: the \$5M lands in a separate account (**BETA, U28113440**), reserves ~\$1.855M for tax, and deploys **slots-first** — ~\$1.8–1.9M directly into the 32 pre-ruled TLH-core slots (single names disperse; a fresh index lot cannot produce 2026 losses by construction), with the **index residual (~\$1.2–1.4M)** as a permanent completion sleeve whose job is beta, not losses. A **Dec-2027 XND collar** hedges the household’s AI concentration on \$1256 terms, phased with deployment. Harvesting runs on two rotation rules (harvest-driven and gate-driven), a **harvest-pair map** built before the money arrives, and a **Nov-28 doubling-up deadline**. The court book (**ALPHA, U24485312**) never mixes: a courted name may never enter BETA; a replication name may never be sized by view.

2. Objectives and binding constraints

#	Constraint	Consequence in design
C1	Losses must realize by 2026-12-31	Slots-first deployment (dispersion now); index residual accepts ~zero 2026 harvest
C2	Tax reserve ≈ \$1.855M (true-up at landing)	SGOV/bills ladder extension day 0; deployable ≈ \$3.2M
C3	Household wash surface = Parametric singles + GOOGL (+ now BETA basket)	Completion-set construction (\$4.3) + wash sweep before every rotation batch
C4	Household already ~26% AI-complex (\$5.18M)	No QQQ; broad index only; XND hedge sized to the <i>overlap</i> , not the sleeve
C5	Judgment/rule boundary (principal-ratified)	Two accounts; courted names structurally excluded from BETA
C6	Ops maturity (this week’s evidence)	No automated order paths in BETA — principal-clicked BasketTrader CSVs only

3. Account architecture

HOUSEHOLD WASH PERIMETER (one taxpayer – rules apply ACROSS ac		
PARAMETRIC 038CAG0E4	ALPHA U24485312 (LIVE)	BETA U28113440 (pending

\$9.19M, 131/31 L/S optimizer-run TLH ~\$105k/mo losses banked (external manager)	court book / hedge fund courted names, envelopes, options templates judgment, gated by courts	\$5M TLH deployment 32 slots + index residua + XND collar + SG0V rese mechanical, no courts
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weekly bundle → exclusion stack ← positions/orders cache
(beta_basket_generator: BETA holds ONLY what
neither Parametric nor ALPHA touches)

- **The dividing line is judgment vs rule, not single-name vs fund.** ALPHA holds anything a court ruled; BETA holds only mechanical replication. This is what makes the wash surface *auditable*: BETA cannot collide with ALPHA by construction, and it cannot collide with Parametric because the basket is built as the **completion set** (index minus Parametric's latest bundle, refreshed weekly).
- **Account pinning:** `require_alpha()` asserts the account on **all four** ALPHA placement paths (`sanyu_accumulator`, `jp_accum_core`, `japan_stager`, `order_executor`) after the Error-435 incident proved an accountless order dies loudly once two accounts exist. BETA has **no automated paths at all** — CSV staged, principal clicks.
- Cash-only crossings between accounts; BETA activation checklist (managedAccounts verification, MCP account-selection check, PA benchmark) lives in `accounts.json`.

4. Sleeve design

4.1 Lane priority (amended 2026-08-21): SLOTS FIRST

1. **Weeks 1–3 post-landing:** deploy ~\$1.8–1.9M directly into the 32 pre-ruled slots (lots 2–3 of the 8/02 map; no new courts — ruled). Staged limit ladders at/near market per \$TINA; envelope-chunked for thin names. Every week earlier = a week more dispersion before 12/31.
2. **Index residual ~\$1.2–1.4M**, same week, in **two non-identical broad funds ~50/50** (VTI + SPLG) so future index-level harvests can swap wash-free. Its job is beta.
3. **Completion criterion:** whatever index remains unrotated at month 12 (floor ~\$0.5M) is the completion sleeve, permanently.

4.2 Rotation rules (the harvest engine)

- **R1 (harvest-driven):** any session with a harvestable index loss $\geq \$5k$ OR any $\geq -1.5\%$ index day → sell index lots (specific-lot, highest basis), buy slot names at their bands in slot-target sizes.
- **R2 (gate-driven):** when a slot band or court gate fires, rotate regardless of index P&L — **court gates outrank harvest timing**. Specific-lot selection minimizes realized gains.
- **Re-base rule:** HELD/STARTER verdicts re-size to ruled percentages on the \$6M base ($0.25\% = \$15k$) through existing gates only; Japan/thin envelopes keep absolute share targets (ADV caps dominate percentage math).

4.3 The basket generator (`desk/beta_basket_generator.py`)

Mechanical, self-documenting, and **any manual edit to its output is a doctrine violation**.
Exclusion stack (union): Parametric latest bundle (`msprime.db`) → research-ledger tickers (every screened name, any state) → edge-classification records → ALPHA live positions → ALPHA resting-order symbols → GOOGL (asserted, never assumed). Output: BasketTrader CSV (LMT at snapshot,

DAY) + manifest JSON. Dry run 8/21: **149 names / \$1.579M**. Refresh cadence: weekly against the newest Bundle_*.zip.

4.4 Harvest-pair map (the December bottleneck — build NOW, pre-landing)

Every slot name gets a designated **non-substantially-identical replacement partner** holding the factor through the 31-day wash window. *No pair, no harvest*. Deadline logic: **Nov 28 is the last doubling-up date** (buy the replacement 31 days before selling the loser to never leave the exposure); names down >8% in early November are doubling-up candidates.

4.5 PROPOSED, NOT RATIFIED — the diligenced-dispersion sleeve

Principal question 8/26: use FLAT-ruled names as the TLH base with paired shorts. Desk analysis: - **Adopted-in-principle half:** flat names are diligenced and carry 40–60% single-name vols → an est. **2–3× harvest yield per dollar** vs the index residual (\$700k–1.2M vs \$350–500k modeled year-one). - **Rejected half:** (a) flat names are *courted* names — holding them in the TLH base reverses the C5 boundary and manufactures cross-account wash collisions exactly where court gates are dated (CYD 8/27, USNA 10/21, AAP 11/12, PWP band); (b) paired single-name shorts are tax-poison in a TLH vehicle (short gains are always ST; several bear-lean flats are MEME-excluded from shorting; shorting a cohort our own study shows drifting +3% fights our own data); (c) duplicates Parametric's L/S engine with worse ops. - **Bounded alternative on the table:** \$750k–1M **long-only** sleeve of flat names passing three filters — liquid at desk rails, not RP_TAINTED, **no live gate inside the harvest horizon** — hedged at book level by the existing XND collar. **Status: awaiting principal ratification; a filter-screen + harvest simulation is the next artifact if pursued.**

5. Hedge architecture (Lane H)

- **Instrument:** XND (NDX-class, ~\$29k/contract) Dec-2027 puts — tenor covers the frozen AI-break date (2027-12-31, p=0.45, CRWV canary). **\$1256 60/40; never QQQ equity options.**
- **Sizing:** household AI/tech overlap ≈ \$4.5M (Parametric \$3.5M + new index tech weight ~\$1.0M); hedge 25–50% → \$1.1–2.2M notional ≈ 38–75 contracts, ~10% OTM. Cost anchor ~4.2%/yr naked → **collared to ~half** by selling calls struck off the **forward** (+8%/16mo), never spot.
- **Phase-in (conditioned-insurance rule):** 1/3 at 50% deployed; 2/3 at full deployment; final 1/3 only if the quarterly AI-break re-freeze holds ≥0.45.
- **Harvest-hedge note:** the tax reserve itself is anti-correlated with the market (a drawdown that hurts the book shrinks the remaining gain to offset) — reserve sits in bills, never risk.

6. Tax accounting rails

- **ST/LT clocks on every lot; tax-adjust every exit** (ARX precedent). Specific-lot ID everywhere.
- **Wash discipline:** sweep vs Parametric + GOOGL + BETA before every rotation batch; the pair map (§4.4) is the standing answer; cross-account collisions are structural-impossible for BETA by the exclusion stack, and for ALPHA by the no-courted-names-in-BETA rule.
- **Parametric interplay:** ~\$105k/mo realized-loss run-rate already banked (–\$425k inception→8/02) — the household's 2026 offset is Parametric + BETA slots combined; **always dedup Parametric G/L by (taxlot, date, qty)** (the 2.5× app inflation lesson).

- **Reserve math:** \$5M → ~\$1.855M owed → ~\$3.2M deployable; true-up at landing against the estimated-payment calendar; >±\$300k movement re-opens the runbook.

7. Infrastructure inventory (what exists, what watches it)

Component	File / agent	Job	Failure mode it closes
Basket generator	desk/beta_basket_generator.py	completion of basket, exclusion stack, CSV	wash-by-construction; judgment leakage
Account registry	desk/account_registry.py (require_alpha)	pipes every automated order to ALPHA	Error-435 / wrong-account placement
Order sentinel	desk/order_sentinel.py + launchd 30-min	pymbie cancels, PendingCancel SLA 30m, cancel-by enforcement , missing-order verify, gateway 2-strike alarm, runner liveness	the 18h stuck-cancel class; resting-into-print rail now machine-held
Intent ledger	desk/data/order_intent_ledger.py	every sentinel/cb directive gets a machine-checked row	directives enforced only by memory
Band watch	desk/band_watch.py	ledger bands vs live tape; order-coverage derived from the blotter each run	suppressed alerts over dead orders (ONON class)
Consistency check	desk/consistency_checker.py	daily dated, resting-orders-have-catalysts, grader health, unverified-ledger drain stall , pre-mortem presence	silent-accumulation and dateless-gate classes
Envelope runner	desk.envelope_runner (Rung 0.75, principal-run + caffeine)	ALPHA Japan accumulators; never BETA	daemon-sleep session loss (sentinel now warns)
Court conveyor	desk.court_runner (hourly launchd)	produces/updates the slot map ALPHA-side; gates that outrank harvest (R2)	—
Mailer / dashboard	desk/mailer.py, desk/UI/ticker/{sym}	every event legible; every ticker → dossier	decisions trapped in chat

Known open infra tickets: court-runner IBKR MCP grant (benches permission-denied — repeated); bench tape-divergence preflight (3 stale-tape instances this docket); runner red/blue pairing defect (3rd recurrence); gateway auto-restart (needs principal security decision); envelope-runner launchd migration (autonomy-rung decision); ZAR permission if MTN.JO ever stages.

8. Operating cadences

- **Daily (deployment window):** rotation-rule scan at close; sentinel every 30m; band watch 6:15/7:45.
- **Weekly:** Parametric bundle refresh → basket regen → drift check; wash sweep.
- **Monthly:** migration pace vs 12/31 runway; hedge phase vs deployment %; reserve true-up; MR-scorecard cohort study.
- **Quarterly:** AI-break re-freeze (gates hedge tranche 3); \$ALPHA-MIGRATION gate review.
- **Hard dates:** landing (~Sept); **Nov 28** doubling-up close; **Dec 31** harvest deadline; month-12 completion criterion.

9. Failure modes and their guards (all OBSERVED this cycle, none hypothetical)

Failure	Instance	Guard now standing
Cancels stuck through gateway restart	ACEL/6626/CRM, 18h PendingCancel	sentinel SLA + re-issue protocol
Gateway wedge, silent	twice, 8/24–25	2-strike CRITICAL alarm
Daemon dies on battery sleep	Tokyo session lost 8/17	caffeinate + plugged-in rule + liveness warn
Hand-maintained state rots	ORDER_COVERED 12/15 stale; ONON through its band	live-blotter derivation; static = fallback only
Dateless gates invisible	916 packs; BGSI 12 days	hourly pack-date guard
Work queue becomes archive	unverified ledger stalled 5 days	drain-stall flag; R1.13(e) scope
Vendor tape lies	PTS marks; intraday-vs-close; BYND reverse split (+2538% fake)	artifact-family memory + split guard in scorecard
Wrong-account order	Error 435	require_alpha in all four paths; BETA has no automated paths

10. Decision queue (principal)

1. Ratify or decline the **bounded dispersion sleeve** (§4.5) — if yes, the three-filter screen + harvest simulation is the next artifact.
2. **§ALPHA-MIGRATION gates** into the runbook (cohort grade 11/22 + Brier record; 30 clean days; \$30–60k slot scale test).
3. Gateway auto-restart (credential/2FA trade-off) and envelope-runner launchd (rung change).
4. BETA activation checklist on account approval (managedAccounts, MCP account selection, PA benchmark).

Prepared by the desk, 2026-08-26. Sources: DEPLOYMENT_RUNBOOK_5M_20260821.md (+8/21 amendments), accounts.json, beta_basket_generator.py, this week's hardening commits, cohort study 8/26.